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Dedicated to Enterprise Migration

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Agenda

1. Common reasons for migration
2. Migration checklist
3. Migration Watch-outs
4. Migration path
5. Estimating costs

Common driving factors

- Costs
  - Dedicated clusters are pricier than Enterprise clusters
  - [Internal] Enterprise clusters have a bigger margin than Dedicated Clusters, allowing us to offer higher discounts if need be
- Spiky workloads
  - Dedicated clusters are pre provisioned. Users need to manually scale the clusters up/down during peaks/lows. Enterprise clusters are serverless
  - Note that this only matters when the peak/low workload goes above CKU limits. It does not make a difference otherwise
- [AWS Only]High networking costs
  - With the introduction of PNI, customers can now reduce their overall networking costs, by migrating over to ESKU

Migration checklist

- The customer is okay deploying a multi-tenant environment
- The customer is not using features missing in the ESKU column here
- [If peering/TGW is being used] The customer is aware of/okay moving to a PL/PNI based networking setup
  - This especially matter for connectors since connector networking over PL is vastly different from Peering based connections
- The customer is not using KSQL
  - If they are, they should be okay moving to Flink

Migration Watch-outs

- **Networking / DNS**
  - Enterprise: use Ingress PL Gateways + Access Points (per-AP endpoints for Kafka/Flink/SR), not legacy PLATT URLs.
  - Dedicated: uses CCN (PL/Peering/TGW) + CCN routing; Enterprise is serverless over PL/AP.
  - Validate VPC/VNet reachability to Kafka/Flink/SR per Access Point; design private hosted zones for the new AP DNS patterns.
  - Ensure Terraform provider ≥ v2.60 and use confluent_gateway, confluent_access_point, confluent_endpoint.

- **Flink**
  - Confirm Flink compute pools can reach Enterprise Kafka + SR over PL/AP.
  - Check cloud/region GA status for Flink private networking + multi-env PL before committing migration dates.

Migration Watch-outs

- **Tableflow**
  - Typical path: Kafka → Flink → Tableflow → Iceberg/S3; Tableflow still reaches SR via public endpoint (by design).
  - Watch per-topic pricing and BYOK on Enterprise for security-sensitive customers.
  - Plan end-to-end schema evolution (Kafka/SR/Flink/Tableflow/Iceberg) to avoid breaking writes.
- **Schema Registry**
  - Dedicated + CCN: SR private endpoint is auto-created; migrate app clients from public → private SR URL.
  - Enterprise: use Access-Point–specific SR endpoints; update app SR URLs + IP filters, knowing internal components (Flink, connectors, Tableflow, ksqlDB) still use public SR and ignore SR IP filters.

Migration Path

1. Discovery & sizing

1. Capture current Dedicated usage:
   - Peak **ingress/egress MBps**, **partition count**, **storage**, and **cluster load metric**.
2. Choose SLA → min eCKU
   - Enterprise supports:
   - 1 eCKU minimum at 99.9% SLA (good for non-prod / cost-sensitive).
   - 2+ eCKU minimum at 99.99% SLA (recommended for prod).
   - This minimum is the floor; Enterprise auto-scales above that as workload grows (you don't pre-size like CKUs).

2. Provision the Enterprise cluster (with Terraform or UI)

1. Use **Confluent Terraform Provider** to create:
   - Environment (if needed),
   - **Enterprise cluster** with required `ecku`/SLA,
   - Private networking (PL/PNI/Peering/PSC) matching your pattern.
2. Reuse or create:
   - Service accounts, RBAC, ACLs,
   - Schema Registry private endpoints (via CCN/PLATT if you already have Dedicated private networking).

3. Establish Cluster Linking (data plane)

1. From Enterprise (destination), create a **cluster link** from the Dedicated source:
   - Use **private-networking–supported combo** Dedicated↔Enterprise in same org.
2. Configure:
   - **Mirror topics** (all or selected),
   - **Consumer offset sync**.
3. Let mirrors **backfill and stay live**; monitor link lag until stable.

4. Migrate consumers & producers

1. **Pilot cutover** per app:
   - Stop a consumer group on Dedicated,
   - Point it to the Enterprise mirror topic and resume.
2. When ready for full cutover:
   - Stop all remaining producers/consumers on Dedicated,
   - Wait for cluster link lag → 0,
   - **Promote mirror topics** on Enterprise to make them writable,
   - Reconfigure all clients to the Enterprise bootstrap & credentials.
3. Keep the link in place briefly in case you need to roll back reads.

5. Migrate Flink jobs

1. For each Flink SQL job / pipeline:
   - Duplicate the tables & jobs pointing to Enterprise topics,
   - Run in **shadow mode** (reading mirrors) to validate results,
   - Switch downstream consumers to use Flink outputs on Enterprise,
   - Decommission Flink jobs that still read from Dedicated.

6. Migrate managed connectors (via Terraform)

6.1 Import existing connector definitions from Dedicated

Choose one:

- Use **Resource Importer** to generate `main.tf` + `terraform.tfstate` for existing connectors, or
- Define `confluent_connector` resources and `terraform import` them.

6.2. Cutover connectors

1. While Cluster Link is mirroring:
   - `terraform apply` the **Enterprise** module instance → creates identical connectors against Enterprise.
2. Validate:
   - Sink connectors: data arriving correctly in targets,
   - Source connectors: topics on Enterprise populated as expected.
3. Once apps and sinks are confirmed on Enterprise:
   - `terraform apply` with Dedicated connectors removed to tear down old ones.

7. Validation, tuning, and decommission

1. Monitor **Enterprise cluster load / eCKU utilization**.
2. Confirm:
   - All apps, Flink jobs, connectors, and schemas use Enterprise endpoints.
3. When comfortable:
   - Shrink or delete the Dedicated cluster (per CKU resize/deletion rules).

Estimating costs

- Good news: Throughput limits per CKU/ECKU are the same; You can just clone their org usage, switch cluster type to Dedicated, and you should be good to go
- General guidance:
  - Create an estimator based off of their last 3 months usage
  - Match the throughput to their average throughput
  - Match the SLA:
    - Remember, ESKU are multitenant by default with 99.9% SLA.
      - If their Dedicated cluster is running on 1CKU, tune the enterprise cluster estimate to run on 99.9%
    - If their cluster is on 2CKU+(and multi AZ), just tune the estimate to 99.99% SLA

Variables to think of

- Heavily provisioned clusters
  - Most Dedicated clusters run on 1-2CKU. If they have provisioned more than this, its either due to partitions or throughput. Handle these scenarios based on the below mentioned guidelines
- Partitions
  - ESKUs come with 1500 less partitions per ECKU as compared to a Dedicated CKU
  - Match the partition count so as to size the ECKUs correctly
- Throughput
  - If the cluster has >2CKU, and the throughput really is high, throughput that needs that many CKUs, use that throughput in your ESKU estimate
  - If the cluster has >2CKU, but the average throughput is below the throughput threshold for 2CKU, understand the reason behind it, and get an estimate on what the average throughput actually looks like

Edge cases

- Highly spiky workloads:
  - For workloads that are very spiky in nature, where the peak throughput needs an additional CKU, understand how often and how long those peaks are observed
    - If peaks are frequent(daily) and last long(hours), your average throughput input in the should be somewhere close to the peak throughput to avoid under sizing them
    - If the peaks are not frequent/don’t last too long, the average throughput is good enough, since the peaks don’t add on to the costs too much

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